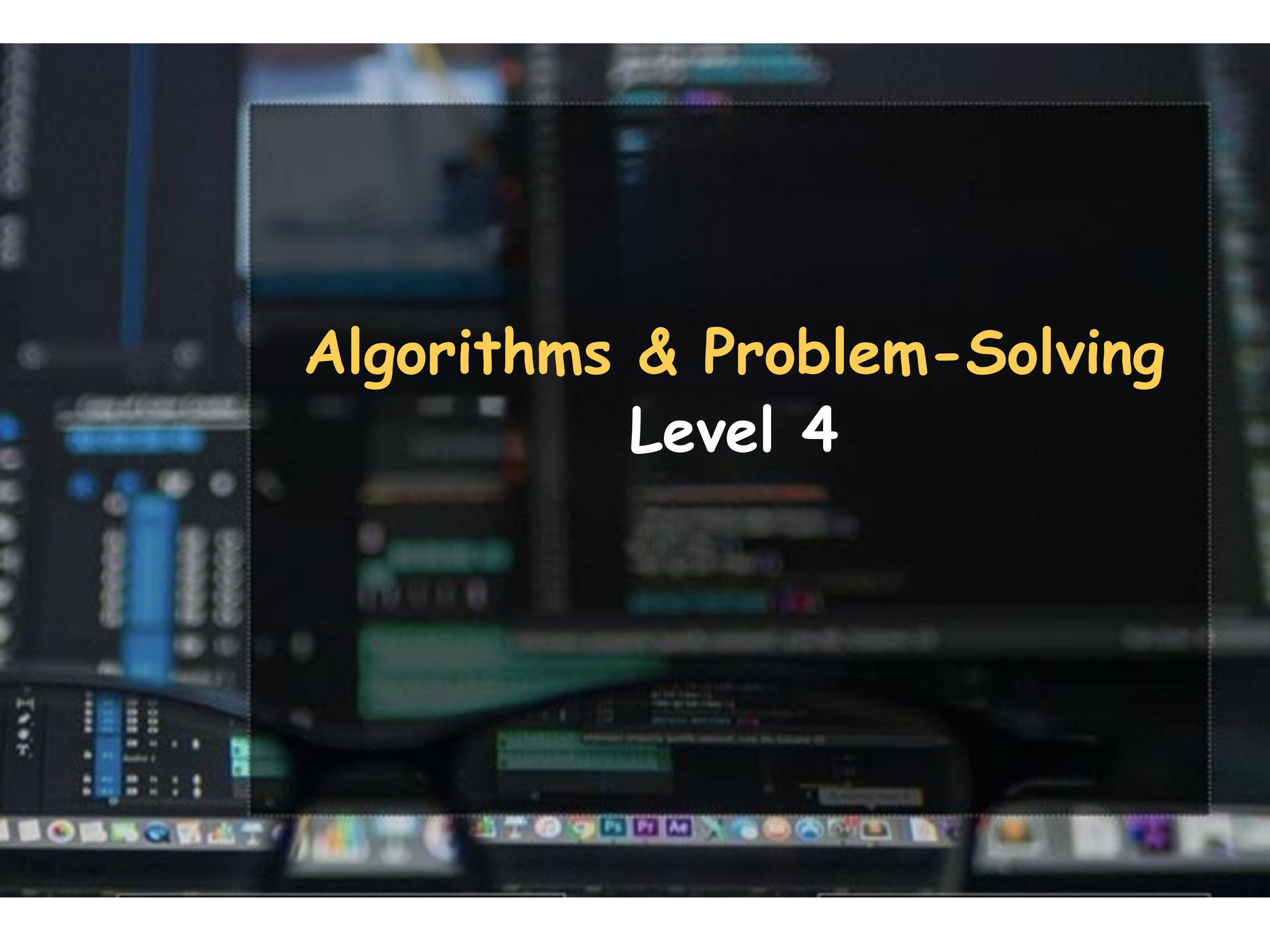
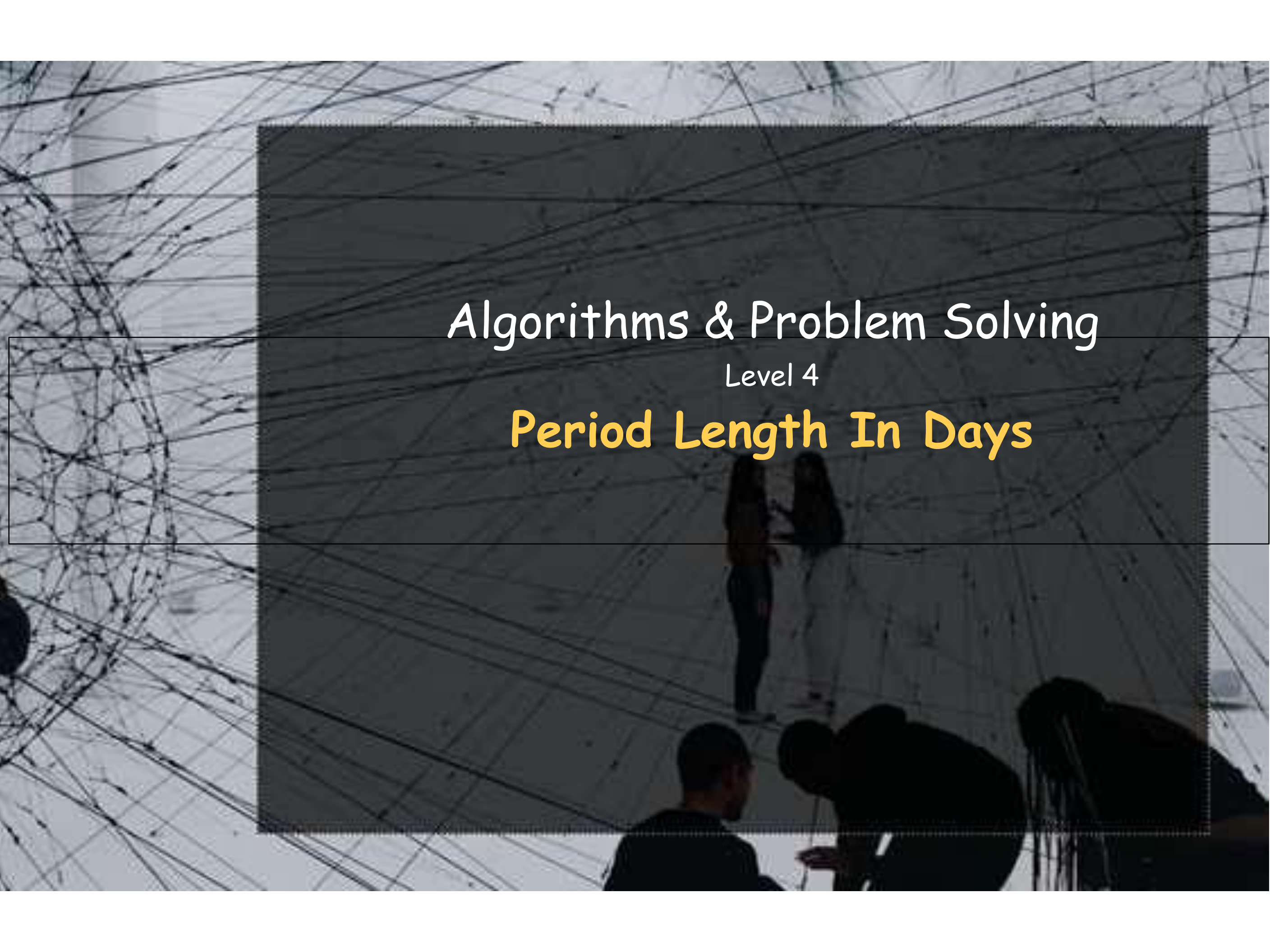




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a group, which adds a human element to the technical-looking background.

Algorithms & Problem Solving

Level 4

Period Length In Days



Definition

- sDate Struct: Represents a single date containing the day, month, and year.
- sPeriod Struct: Represents a time period between two dates:
 - StartDate - Start date
 - EndDate - End date



Explanation:

- This program reads a time period (start date and end date) and then calculates the number of days between them, with the option to include the end day within the period or exclude it.
- `IsLeapYear(year)` Checks if a year is a leap year (i.e., has 29 days in February)
- `NumberOfDaysInAMonth(month, year)` Returns the number of days in a given month, taking leap years into account
- `IncreaseDateByOneDay(date)` Increments the date by one day, moving to the next month or year if needed
- `GetDifferenceInDays(date1, date2, includeEndDay)` Calculates the number of days between two dates by repeatedly incrementing the date until the second date is reached
 - If `includeEndDay = true`, the end date is also calculated
- `PeriodLengthInDays(period, includeEndDate)` Calculates the length of a period based on the start and end dates

Procedures

- 1 Read the start date of the period
- 2 Read the end date of the period
- 3 Calculate the difference between the two dates using `PeriodLengthInDays()`
- 4 Print the period twice:
 -  Once without including the end date
 -  Once with including the end date

Example:

Start Date: 1/1/2022

End Date: 5/1/2022

Period Length is: 4

Period Length (Including End Date) is: 5

Explanation:

- The difference between day 1 and day 5 is calculated as follows:
 $1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 4, 4 \rightarrow 5 = 4$ days excluding the final day. However, if we include the final day, it becomes 5 days.



Note:

- The program does not verify the correct date order, so the start date must be manually placed before the end date.

Summary

- ✓ The program accurately calculates the length of the time period, even across months and years.
- ✓ It supports the option to calculate the duration with or without an end date.
- ✓ Useful for time management, vacation, or billing applications.

Thank you 😊